

Small-world properties of the Indian Railway network

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Structural properties of the Indian Railway network is studied in the light of recent investigations of the scaling properties of different complex networks. Stations are considered as ‘nodes’ and an arbitrary pair of stations is said to be connected by a ‘link’ when at least one train stops at both stations. Rigorous analysis of the existing data shows that the Indian Railway network displays small-world properties. We define and estimate several other quantities associated with this network.

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Given a chance, how would we have possibly organized our train travel? People dislike to change trains to reach their destinations. Therefore an extreme possibility would be to run a single train passing through all stations in the country so that no change of train is needed at all! An obvious disadvantage in this strategy is that the average distance between the stations become very large and so also the time needed for travel. The other limiting situation would be, to run a train between any pair of neighbouring stations and try to travel along the minimal paths. This requires a change of train at every station, which is also clearly not economically viable. Railway networks in no country in the world follow either of the two ways, actually they go mid-way. Like any other transport system the main motivation of railways is to be fast and economic. To achieve it, railways run simultaneously many trains, covering short as well as long routes so that a traveller does not need to change more than only a few trains to reach any arbitrary destination in the country.

In this paper we analyse the structure of the Indian Railway network (IRN). This is done in the context of recent investigations of the scaling properties of several complex networks e.g., social, biological, computational networks [1] etc. Identifying the stations as nodes of the network and a train which stops at any two stations as the link between the nodes we measure the average distance between an arbitrary pair of stations and find that it depends only logarithmically on the total number of stations in the country. While from the network point of view this implies the small-world nature of the railway network, in practice a traveller has to change only few trains to reach an arbitrary destination. This implies that over years, the railway network has been evolved with the sole aim in mind to make it fast and economic, eventually its structure has become a small-world network [2].

The structure and properties of several social, biological and computational networks like the World-wide web (WWW) [3], network of the Internet structure [4], neural networks [5], collaboration network [6] etc. are being

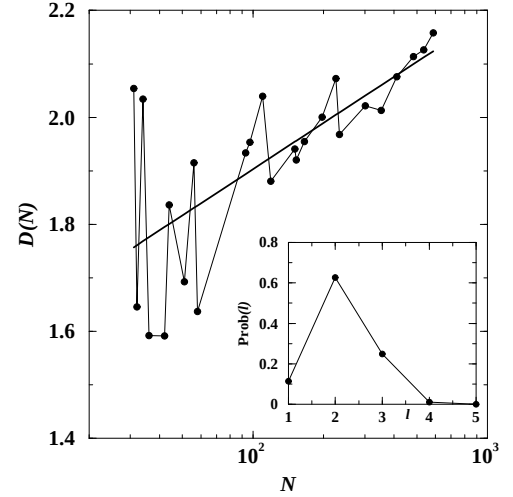


FIG. 1: The variation of the mean distance $D(N)$ of 25 different subsets of IRN having different number of nodes (N). The whole range is fitted with a function like $D(N) = A + B \log(N)$ where $A \approx 1.33$ and $B \approx 0.13$. The inset shows the distribution $\text{Prob}(\ell)$ of the shortest path lengths ℓ on IRN. The lengths varied to a maximum of only five link lengths and the network has a mean distance $D(N) \approx 2.16$.

studied recently with much interest. In general a network has a number of ‘nodes’ and some ‘links’ connecting different pairs of nodes. Typically the following quantities are defined to characterize a network of N nodes: (i) the diameter is the maximum distance between an arbitrary pair of nodes (ii) the clustering coefficient $\mathcal{C}(N)$ is the average fraction of connected triplets (iii) the probability distribution $P(k)$ that an arbitrarily selected node has the degree k i.e. this node is linked to k other nodes.

Watts and Strogatz [2] proposed a model of small-world network (SWN) in the context of various social and biological networks. They argued that SWNs must have small diameters which grow as $\ln N$ like random networks but should have large values of the clustering coefficients $\mathcal{C}(N) \sim 1$ like regular networks. On the other

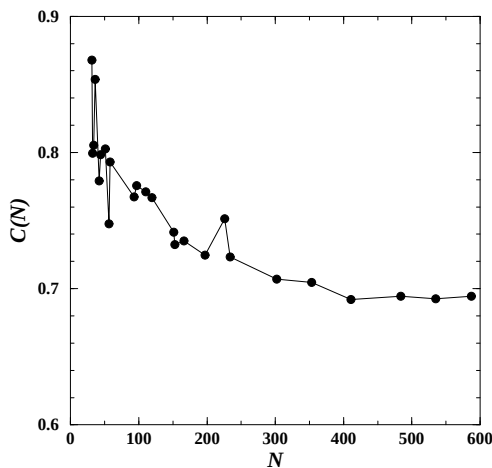


FIG. 2: Variation of the clustering coefficient $\mathcal{C}(N)$ of 25 different subsets of IRN having different number of nodes N . Starting from a somewhat higher value at small number of nodes, the clustering coefficient decreases slowly on increasing N and finally saturates at 0.69.

hand the scale-free networks (SFN) are characterized by the power law decay of the degree distribution function: $P(k) \sim k^{-\gamma}$. It was observed later that the degree distributions of nodes for two very important networks e.g., World-wide web [3] which is a network of web-pages and the hyperlinks among various pages and the Internet network [4] of routers or autonomous systems have scale-free property. Barabási and Albert (BA) proposed a model for SFN which grows from an initial set of nodes and at every time step some additional nodes are introduced which are randomly connected to the previous nodes with the linear attachment probabilities [7]. All scale-free networks are believed to display small-world properties while a small-world network is not necessarily scale-free.

Networks defined on the Euclidean space have also generated much interests in recent times. Internet, transport systems, postal networks etc. are naturally defined on two-dimensional space. In these generalised networks the attachment probabilities depend jointly on the nodal degrees as well as the lengths of the links [8, 9].

A railway network is one of the most important examples of transport systems. The very complex topological structures of railway networks have attracted the attention of researchers in many different contexts. For example the fractal nature of the structure of railway networks was studied by Benguigui [10]. Very recently the efficiency of Boston subway network has been studied where a new measure for such networks has been proposed [11].

Our scheme is to associate first a representative graph G_N with the IRN of N stations in the following way. Here the stations represent the ‘nodes’ of the graph, whereas two arbitrary stations are considered to be connected by a ‘link’ when there is at least one train which stops at

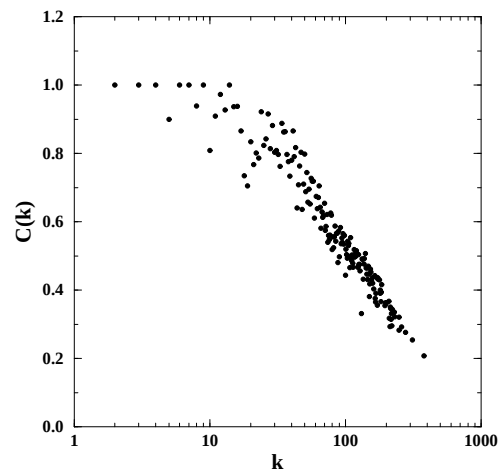


FIG. 3: The variation of the clustering coefficient $\mathcal{C}(k)$ against the degree k for the IRN indicates a logarithmic decay for large k .

both the stations. These two stations are considered to be at unit distance of separation irrespective of the geographical distance between them. Therefore the shortest distance ℓ_{ij} between an arbitrary pair of stations s_i and s_j is the minimum number of different trains one needs to board on to travel from s_i to s_j . Thus $\ell_{ij} = 1$ implies that there is at least one train which stops at both s_i and s_j . Similarly, $\ell_{ij} = 2$ implies that there is no train which stops at both s_i and s_j and one has to change the train at least once in some intermediate station to board the second train to reach s_j . With this definition, if the trains $t_1, t_2, \dots, t_n$ pass through a station s_i , then all the stations through which these n trains pass are unit distance away from s_i and are considered as first neighbours of s_i . Consequently, the number k_i of such stations is the degree of the node s_i .

Indian Railway network is a densely populated network of more than 8000 stations where the number of trains plying in this network is of the order of 10000 [12]. However, we collected the data of IRN on a coarse-grained level following the recent Indian Railways time table ‘Trains at a Glance’ [13] containing the important trains and stations in India. This table contains a total of $L = 579$ trains covering $N = 587$ stations in a total of 86 tables. A grand rectangular matrix $\mathcal{G}(N, L)$ is then constructed such that the ij -th element of this matrix is 1 if the train j stops at the station i , otherwise this element is zero. A second matrix $\mathcal{T}(0 : N, N)$ is also constructed where the degree k_i of the station i is stored at the element $\mathcal{T}(0, i)$ and the serial numbers of the k_i neighbours of i are stored at the locations $\mathcal{T}(j, i), j = 1, k_i$, rest of the elements being zero. We define and estimate the following quantities for the IRN.

Since G_N is a connected graph, there are $N(N - 1)/2$

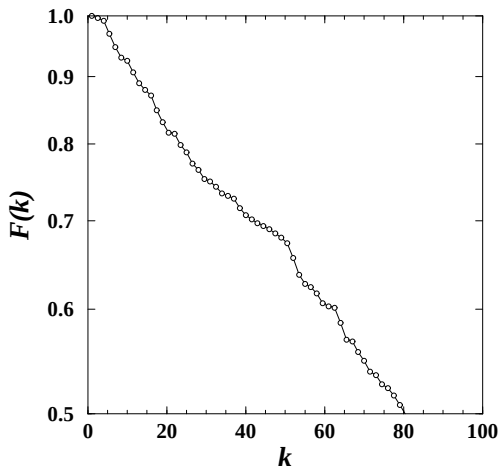


FIG. 4: The cumulative degree distribution $F(k)$ of the IRN with the degree k is plotted on the semi-logarithmic scale.

distinct shortest paths among the N stations. We calculate the probability distribution of the shortest path lengths $\text{Prob}(\ell)$. The shortest path lengths are calculated using a burning algorithm [14] and using the matrix $\mathcal{T}$. In this algorithm the fire starts from an arbitrary node i , and burns this node at time $t = 0$. At time $t = 1$ the fire burns all k_i neighbours of i . At time $t = 2$ all unburnt neighbours of k_i nodes are burnt and so on. The burning time of a node is the length of the shortest path of that node from the node i . This calculation has been repeated for all N nodes to get $N(N-1)/2$ shortest distances. In Fig. 1 inset we plot this distribution which goes to a maximum of $\ell = 5$ implying that one needs to change at most four trains to reach any station from any station in India on the coarse-grained level. Similarly the distribution has a peak at $\ell = 2$ implying that one can go to the majority of stations in India by changing train only once. In the graph theory the diameter of a graph is measured by the maximum distance between the pairs of nodes. Therefore according to this definition the diameter of our network is exactly equal to 5. However the average shortest path between an arbitrarily selected pair of nodes which we call as the mean distance $\mathcal{D}(N)$ is also a measure of the topological size of the graph and have been used by many authors to measure the size of networks as described in [7]. We therefore measure the mean distance $\mathcal{D}(N)$ of the railway network of N stations as the average shortest distance $\langle \ell_{ij} \rangle$ between an arbitrary pair of stations s_i and s_j . We obtain $\mathcal{D}(N) \approx 2.16$ for this network.

It is desirable to see how $\mathcal{D}(N)$ varies with N [15]. Since we have the data of a single railway network, we divide the whole IRN into 25 different subsets consisting of trains and stations of 10 different states, 7 different combinations of states, 7 different railway zones and

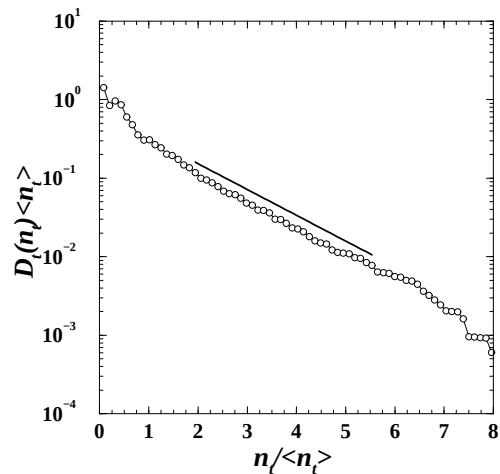


FIG. 5: Scaled probability distribution $D_t(n_t)$ for an arbitrary station through which n_t trains pass where $\langle n_t \rangle \approx 12.06$. Binned data is presented through the circles connected by lines which fits best to an exponential form: $D_t(n_t)\langle n_t \rangle = a \exp(-bx)$ with $x = n_t/\langle n_t \rangle$, $a \approx 0.47$ and $b \approx 0.75$.

the whole IRN. As a result we obtained 25 data points (though they are not necessarily non-overlapping samples), reflecting the nature of variation of $\mathcal{D}(N)$ with N . In Fig. 1 we plot this data on a semi-log scale and though there is some wild fluctuations for small values of N , for large values of N the linear behaviour is quite apparent. The whole range is fitted with $\mathcal{D}(N) = A + B \log(N)$ where $A \approx 1.33$ and $B \approx 0.13$.

The clustering coefficient $\mathcal{C}(N)$ is defined in the following way. Let the subgraph G_i consisting of the neighbours of s_i i.e., $(s_1, s_2, s_3, \dots, s_{k_i})$ have E_i links among them. Then the clustering co-efficient $\mathcal{C}_i$ of the node i is $2E_i/k_i(k_i-1)$ and that of the whole network is $\mathcal{C}_i = \langle \mathcal{C}_i \rangle$. A direct measure of the clustering co-efficient of the whole IRN gives: $\mathcal{C} \approx 0.69$ (Fig. 2). The high value of the clustering coefficient is explained in the following way. The number n_s of stations in which a particular train stops are all at unit distance from one another on the network and therefore form an n_s -clique. Therefore if only one train stops at some station i then $\mathcal{C}_i = 1$. When two trains stop at the station i and the sets $n_s(1)$ and $n_s(2)$ of stations covered by these two trains are different, $\mathcal{C}_i$ is in general smaller than 1. However there may be other trains which do not stop at i but stop at the stations which are not in both $n_s(1)$ and $n_s(2)$. These trains enhance the value of $\mathcal{C}_i$. The value of $\mathcal{C} \approx 0.69$ is compared with a corresponding random graph network having the same number of vertices and edges as in IRN with the edges distributed randomly. It is found that the number of edges in IRN is 19603. If these edges are distributed randomly within the maximum possible edges on a graph of $N=587$ nodes the clustering coefficient should be $19603/[N(N-1)/2] \approx 0.113$ which is the same

as Prob(1). We also compute a modified clustering coefficient $\mathcal{C}_o$ by counting in E_i only those links in the sub-graph G_i which pass through the node i . We obtained a value $\mathcal{C}_o \approx 0.55$ for the IRN.

Recently, the study of the clustering coefficient as a function of the degree of the node of some real-world network has shown an interesting feature [17] $\mathcal{C}(k)$, defined as the clustering coefficient of the node with degree k , shows a decrease (apparently a power law decay) with k in several networks like the actor, language or world-wide-web networks. However in the network of internet at the router level or power grid network of the Western US, $\mathcal{C}(k)$ was found to be more or less a constant. In the IRN also, we find that $\mathcal{C}(k)$ (Fig. 3) remains at a constant value close to unity for small k and shows a logarithmic decay at larger values of k . In all these real-world networks where $\mathcal{C}(k)$ remains more or less a constant, the nodes are linked by physical connections which may be responsible for this common feature. However, in this context it should also be mentioned that the scale-free Barabási-Albert network [7] also predicts $\mathcal{C}(k) \propto k^0$ and $\mathcal{C}(N) \propto N^{-0.75}$. In the IRN, although $\mathcal{C}(N)$ shows a decrease with N , it is apparently much slower than a power law.

The degree distribution of the network, that is, the distribution of the number of stations k which are connected by direct trains to an arbitrary station is denoted by $P(k)$. We plot the cumulative degree distribution $F(k) = \int_k^\infty P(k)dk$ using a semi-log scale in Fig. 4 for the whole IRN. We see that $F(k)$ approximately fits to an exponentially decaying distribution $F(k) \sim \exp(-\alpha k)$ with $\alpha = 0.0085$.

We also calculated the distribution $D(n_t)$ of the number of trains n_t which stop at an arbitrary station. This is plotted in Fig. 5 on a semi-log scale after scaling by the average number of trains $\langle n_t \rangle \approx 12.06$ along both the abscissa and the ordinate. The data is binned as before and is fitted to an exponential form: $D_t(n_t)\langle n_t \rangle = a \exp(-bx)$ with $x = n_t/\langle n_t \rangle$, $a \approx 0.47$ and $b \approx 0.75$.

The distribution $D(n_s)$ of the number of stations through which an arbitrary train passes is plotted in Fig. 6. The data is scaled by the average number of stations $\langle n_s \rangle \approx 12.37$ along both the abscissa and the ordinate. The $D(n_s)$ grows very fast at the beginning, reaches a maximum and then decays to zero. A numerical fit to a functional form like $D_s(n_s)\langle n_s \rangle = ax^4/(x^2 + b)^3$ with $x = n_s/\langle n_s \rangle$, $a \approx 0.6$ and $b \approx 0.096$ turns out to be reasonably good.

We also measure the connectivity correlation of IRN following the works of [16]. Let $F(k'|k)$ denote the conditional probability that a node of degree k has a neighbour of degree k' . Then to see how the nodes of different degrees are correlated we measure the average degree $\langle k_{nn}(k) \rangle = \sum_{k'} k' F(k'|k)$ of the subset of nodes which are all neighbours to a particular node of degree k . In general this average has a variation like $\langle k_{nn}(k) \rangle \sim k^{-\nu}$ where

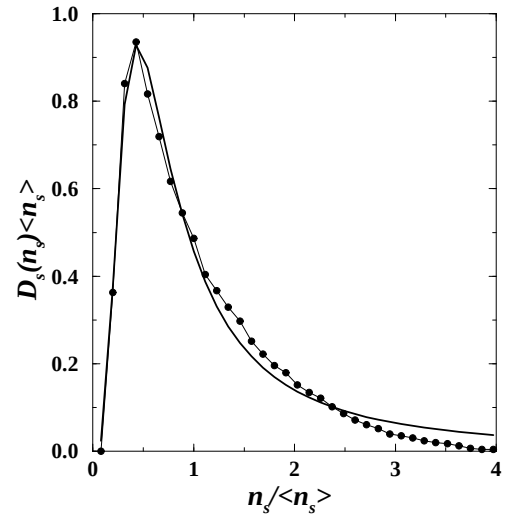


FIG. 6: Scaled probability distribution $D_s(n_s)$ for an arbitrary train passing through n_s stations where $\langle n_s \rangle \approx 12.37$. Binned data is presented through the black dots connected by lines which fits best to the form: $D_s(n_s)\langle n_s \rangle = ax^4/(x^2 + b)^3$ with $x = n_s/\langle n_s \rangle$, $a \approx 0.6$ and $b \approx 0.096$.

a non-zero ν reflects a non-trivial correlation among the nodes of the network. We calculated $\langle k_{nn}(k) \rangle$ for IRN and plotted it in Fig. 7 on a double logarithmic scale. Almost over a decade the $\langle k_{nn}(k) \rangle$ remains same on the average and is independent of k , indicating the absence of correlations among the nodes of different degrees.

A more sensitive measure for the degree correlations was proposed in [18]. Newman has defined a degree-degree correlation function r which measures whether a vertex of high degree at one end of a link prefers a vertex of high degree (“assortative mixing”, $r > 0$) or low degree (“disassortative mixing”, $r < 0$) at the other end. It has been observed that social networks are assortative and technological and biological networks are disassortative. We have measured for IRN the normalized correlation function following [18] and found its values to be $r = -0.033$. This indicates that the IRN is of *disassortative* nature, i.e. rich vertices at one end of a link show some preference towards poor vertices at the other end, and vice versa.

To summarize, we investigated the structural properties of the Indian Railway network to see if some of the general scaling behaviour obtained for many complex networks in recent times may also be present in IRN. While nodes of the network are evidently the stations, the links are defined as the pairs of stations communicated by single trains. With such a definition of link, the mean distance of the network is a measure of how good is the connectivity of the network. Indeed, we observed that the mean distance of IRN varies logarithmically with the number of nodes with a high value of the clustering coef-

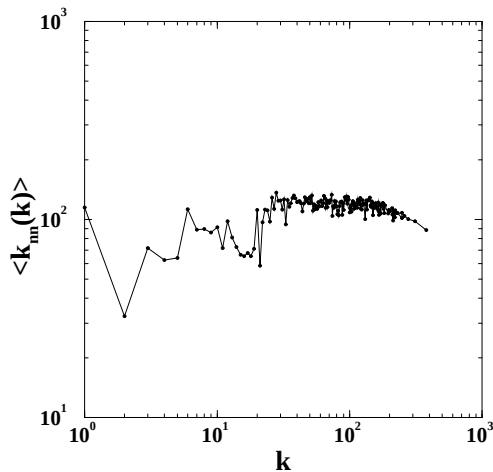


FIG. 7: The variation of the average degree $\langle k_{nn}(k) \rangle$ of the neighbours of a node of degree k with k . After some initial fluctuations, $\langle k_{nn}(k) \rangle$ remains almost same over a decade around $k = 30$ to 300 indicating absence of correlations among the nodes of different degrees.

ficient. This implies that IRN behaves like a small-world network, which we believe should be typical of the railway network of any other country, which we are unable to study at present for unavailability of data.

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